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Chromosomes and Gene Expression

Distal Gene Expression Governed by Lamins and Nesprins via Chromatin Conformation Change

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eLife Assessment

This study provides **useful** information on the impact of Lamin A/C knockdown on gene expression using RNA-Seq analysis. In addition, the impact of Lamin A/C knockdown on telomere dynamics is explored using live cell imaging. The conclusions, however, are **inadequately** supported by the data presented. Weaknesses include excessive reliance on gene ontology analysis without further validation of direct versus indirect effects, use of only one shRNA, which may have off target effects, validation of knockdown only from gene expression rather than protein levels, lack of discussion on previous studies showing the presence of Lamin A/C in the nuclear interior among others.

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Abstract

The nuclear lamina is a vital structural component of eukaryotic cells, playing a pivotal role in both physiological processes, such as cell differentiation, and pathological conditions, including laminopathies and cancer metastasis. Lamina associated proteins, particularly lamins and nesprins, are integral to mechanosensing, chromatin organization, and gene regulation. However, their precise contributions to gene regulation remain incompletely understood. This study explores the functions of lamin A, LMNA, and SYNE2 in gene expression, with a particular focus on their influence on distal chromatin interactions and conformational changes. Using inducible shRNA knockdown, RNA-seq analysis, and dCas9-mediated live imaging of chromosomes, we demonstrate that lamin A affects RNA synthesis, LMNA governs chromatin spatial organization, and SYNE2 regulates chromatin modifications. Furthermore, both lamins and nesprins enhance telomere dynamics. These findings elucidate nuclear envelope-associated mechanisms in gene regulation, offering valuable insights into chromatin dynamics under both physiological and pathological contexts.

1. Introduction

The three-dimensional organization of the genome within the nucleus plays a crucial role in coordinating gene expression, ensuring the stable repression of heterochromatin while permitting the dynamic regulation of transcriptionally active regions. This organization is intimately connected to the nuclear envelope, a specialized membrane enriched with proteins that actively contribute to chromatin architecture and gene regulation[1]. Among these proteins, the nuclear lamina- a filamentous meshwork of intermediate filament (IF) proteins known as lamins—forms a structural scaffold that anchors chromatin to the nuclear periphery, thereby influencing nuclear mechanics and genomic stability [2]. Lamins interact with the linker of nucleoskeleton and cytoskeleton (LINC) complex to form a direct mechanotransduction bridge that links the cytoskeleton to the extracellular matrix (ECM) while simultaneously influencing chromatin topology. A-type lamins, encoded by the *LMNA* gene, include lamin A and lamin C, whereas B-type lamins (*LMNB1* and *LMNB2*) maintain constitutive roles in nuclear structure across different cell types[3–7].

Lamins have been implicated in diverse biological processes, including stem cell differentiation, cancer progression, aging, and laminopathies, underscoring their multifaceted role in nuclear integrity and genome regulation. Nesprins, giant spectrin-repeat proteins of the LINC complex, serve as critical linkers between the cytoskeletal filaments and the nuclear lamina, providing structural integrity and modulating intracellular signaling [8,9].

While significant progress has been made in understanding the role of lamins in genome organization, the precise mechanisms by which lamins and nesprins regulate gene expression through distal chromatin interactions remain incompletely understood. Notably, recent evidence suggests a reciprocal interplay between transcription and chromatin conformation, where gene activity can influence chromatin folding and vice versa[10]. However, whether lamins and nesprins actively govern chromatin remodeling and isoform switching beyond their well-characterized functions in mechanotransduction remains an open question.

Lamina-associated domains (LADs) represent large chromatin regions tethered to the nuclear lamina and are generally transcriptionally repressive, enriched with heterochromatic marks such as H3K9me2/3 and H3K27me3. These domains play a pivotal role in nuclear architecture by segregating active and inactive chromatin compartments. Disruptions in LAD organization have been implicated in diseases such as cancer and laminopathies, highlighting the importance of nuclear envelope components in maintaining genomic stability. However, the extent to which lamins and nesprins regulate distal non-LAD chromatin interactions and isoform switching remain largely unexplored.[11,12]

This study aims to dissect the roles of lamin A, LMNA, and SYNE2 in gene expression and chromatin organization, focusing on their influence on distal chromatin interactions and isoform switching. Using inducible shRNA-mediated knockdown, RNA sequencing (RNA-seq), and dCas9-based live-cell imaging of chromosomes, we systematically investigate their contributions to chromatin conformation and transcriptional regulation. Our findings reveal that lamin A is essential for RNA synthesis, LMNA orchestrates chromatin spatial organization, and SYNE2 modulates chromatin modifications. Furthermore, we demonstrate that both lamins and nesprins impact telomere dynamics, linking nuclear envelope function to chromatin behavior.

By uncovering distinct, isoform-specific and spatially segregated transcriptional responses, our study provides new mechanistic insights into how nuclear lamina components regulate distal gene expression. These findings have broad implications for understanding nuclear architecture in both physiological and pathological contexts, including cancer, aging, and laminopathies.

2. Materials and methods

2.1 Cell culture

Human osteosarcoma (U2OS) cells (#HTB-96, ATCC, USA) and human embryonic kidney (HEK293T) cells (#CRL-3216, ATCC, USA) were cultured in modified Dulbecco's Modified Eagle's Medium (#11965092, DMEM; Gibco), supplemented with 10% fetal bovine serum (FBS). U2OS cells were cultured using HyClone™ characterized tetracycline-screened FBS (#SH30071.03T) at a density of 5×10^4 cells per cm^2 for RNA extraction experiments, and at a density of 1×10^3 cells per cm^2 for Chromosomes live imaging experiments. While HEK293T cells were cultured with Gibco FBS (#A5256701) at an intial density of 5×10^4 cells per cm^2. The medium also contained 100 U ml^{-1} penicillin and 100 µg ml^{-1} streptomycin (#15140122, Gibco). All cultures were maintained in a humidified incubator at 37 °C with 5% CO_2.

2.2 Construction of stable U2OS cells within inducible shRNA plasmids

To generate the doxycycline-inducible knockdown stable cell lines of Scramble, Lamin A, LMNA, SYNE2 shRNAs. targeting shRNAs (Supplementary Table S1) were cloned into the pLKO-Tet-On (#21915, Addgene) inducible vectors by AgeI and EcoRI restriction enzymes (New England Biolabs, USA). HEK293T cells were transfected with pLKO-Tet-On-shRNA, pPAX2 and pMD2 with ratio of 1:1 using Lipofectamine™ 3000 (#L3000015, ThermoFisher Scientific, USA). Lentiviral supernatants were harvested after 48 h, filtered through a 0.45-µm filter, and stored at −80 °C until further use. The Multiplicity of Infection (MOI) was determined by limiting dilution of lentiviral particles. U2OS cells were infected with lentivirus at an MOI of 0.8 in the presence of 8 µg ml^{-1} polybrene (#H9268, Hexadimethrine bromide; MilliporeSigma, USA). Following infection, cells were selected for five days using 4 µg ml^{-1} puromycin (#P4512, MilliporeSigma, USA). Single-cell clones were subsequently isolated and expanded in the presence of 2 µg ml^{-1} puromycin to establish doxycycline-inducible shRNA-knockdown stable cell lines. For doxycycline-inducible shRNA-knockdown stable cell lines, the cells were treated with 100 ng ml^{-1} doxycycline (#D5207, MiliporeSigma, USA) for 48 hours.

2.3 Reverse transcription quantitative real-time PCR (RT-qPCR)

The knockdown efficiency of targeting genes in the stable U2OS cells containing inducible shRNA plasmids was assessed using RT-qPCR. Total RNA was extracted by Trizol (#T3934, MiliporeSigma, USA), 500 ng of total RNA was used as a template for reverse transcription into cDNA using the High-Capacity cDNA Reverse Transcription Kit (#4374966, ThermoFisher Scientific, USA) and real-time PCR was performed using PowerUp™ SYBR™ Green Master Mix (#A25778, ThermoFisher Scientific, USA). For the procedure of real-time PCR, cDNA templates, SYBR, and primers were mixed and loaded to Applied Biosystems™ 7500 Real-Time PCR Systems, started with an initial denaturation at 95 °C for 300 seconds to separate DNA strands. Following 40 cycles of denaturation (95 °C for 20 seconds), annealing (55 °C for 20 seconds), and extension (72 °C for 20 seconds). The relative transcript abundance was initially normalized to GAPDH and subsequently adjusted relative to the no-doxycycline-treated controls using the ΔΔCt method. Primer sequences are shown in Supplementary Table S2.

2.4 Library preparation for transcriptome sequencing (RNA-seq)

Total RNA, including mRNA and lncRNA, was used to prepare sequencing libraries.. RNA was purified from total RNA using probes to remove rRNA by Ribo-Zero rRNA Removal Kit (#15066012, Illumina, USA). RNA integrity and quantification were assessed using the Bioanalyzer 2100 system (Agilent Technologies, USA). RNA was fragmented with divalent cations at high temperature in a

First Strand Synthesis Reaction Buffer (5×). First strand cDNA was synthesized with random hexamer primers and M-MuLV Reverse Transcriptase (RNase H), followed by second strand cDNA Synthesis using DNA Polymerase I and RNase H. The concentration of cDNAs in the library was quantified to 1 ng μl^{-1} using a Qubit 2.0 fluorometer. Overhangs were converted to blunt ends using exonuclease and polymerase activities. After adding adenylated 3' ends to the DNA fragments, NEBNext Adapters with hairpin loops were ligated for hybridization.

To select cDNA fragments 370–420 bp in length, libraries were purified with the AMPure XP system (Beckman Coulter). Size-selected, adaptor-ligated cDNA was treated with 3 µl USER Enzyme (New England Biolabs) at 37 °C for 15 minutes, followed by 95 °C for 5 minutes. PCR amplification was performed using Phusion High-Fidelity DNA Polymerase, Universal PCR primers, and Index (X) primers. The PCR products were purified using the AMPure XP system, and library quality was assessed using the Agilent Fragment Analyzer 5400 system.

Clustering of index-coded samples was carried out on a cBot Cluster Generation System using the TruSeq PE Cluster Kit v3-cBot-HS (Illumina) according to the manufacturer's protocol. Finally, libraries were sequenced on an Illumina Novaseq 6000 platform, generating 150 bp paired-end reads.

2.5 Telomere tracking in live cells by dCas9 imaging

The *S. pyogenes* Cas9 gene, modified with D10A and H840A mutations (dCas9), was engineered to include a 2× SV40 nuclear localization sequence (NLS) and a 3× Flag-tag at the N-terminus, as well as GFP at the C-terminus [13,14]. These dCas9 constructs were cloned into a lentiviral vector driven by the SFFV promoter. To establish a stable dCas9-expressing cell line, cells were transduced with the lentivirus, selected with 5 $\mu g\ ml^{-1}$ puromycin for five days, and subsequently sorted via fluorescence-activated cell sorting (FACS) [14,15]. The sgRNA plasmids were expressed under the U6 promoter, with the original plasmid backbone modified to include TagBFP as a transduction marker. The sgRNA sequences were amplified from an oligo template containing the 20-nucleotide telomere-targeting sequence: GUUAGGGUUAGGGUUAGGGUUA. For viral transduction, cells were incubated with a virus solution diluted threefold in DMEM medium supplemented with 10 $\mu g\ ml^{-1}$ polybrene for 24 hours. For imaging telomere dynamics, fluorescence signals were captured using TIRF microscopy on a Nikon ECLIPSE Ti2 inverted microscope equipped with a Hamamatsu C11440-22C digital CMOS camera. The imaging parameters included an 80° incident angle, 200 ms intervals, and 300 ms exposure time. The laser power was set to 0.8 mW, and the gamma parameter was adjusted to 1[16,17]. At least 20 individual cells were examined for each case in the analysis. Telomere movement was tracked using CellProfiler software (version 4.2.25, *https://cellprofiler.org/*).

2.6 RNA-seq analysis

For trimming, trimming procedure was first applied to eliminate adapter sequences present in our data and to improve read quality from the FASTQ files. Adapter removal and quality trimming were carried out using Trimmomatic (v0.35). In all cases only reads with a Phred quality score > 20 and read length > 50 bp were selected for downstream analysis. Paired-end reads were removed if either read contained more than 10% ambiguous bases (N). Additionally, reads were discarded if more than 50% of their bases had a quality score of ≤ 5. Reads containing adapter sequences were also excluded from further analysis. Each sample produced approximately 80 million sequencing reads after quality control. We then use Salmon (v1.8.0) pseudoalignment to perform alignment, counting and normalization in one single step. The isoform switching was analyzed using the IsoformSwitchAnalyzeR (v1.14.1) [18]. The isoform switches were identified and quantified from RNA-seq data by using gtf annotaion files from gencode (v47). IsoformSwitchAnalyzeR assesses isoform usage by calculating isoform fraction (IF) values, which represent the proportion of a gene's total expression attributed to a specific isoform. This is determined by dividing the expression level of the isoform by the overall gene expression. Changes in isoform usage are quantified as the difference in isoform fraction (dIF), computed as IF2 − IF1. These dIF values are subsequently utilized to determine the isoform log2 fold change.

2.7 Statistical analysis

Each experiment was performed with a minimum of three replicates. All the analysis and comparisons were performed in R (v4.2.0). The DESeq2 (v1.46.0) package was utilized to analyze the differences between the doxycycline-treated groups and the untreated groups [19]. Differential gene analysis was performed with a baseMean threshold of > 50 and a significance level of p-value < 0.05. Benjamini-Hochberg (BH) procedure was employed to control the false discovery rate (FDR) in multiple hypothesis testing (q-value). The resulting differential expressed gene (DEG) lists, ranked in descending order, were subsequently employed for Gene Ontology (GO) enrichment analysis using the clusterProfiler package (v3.20) [20]. For significant levels of multiple group comparison, non-parametric Kruskal-Wallis test was used to compare the medians of multiple groups.

3. Results

3.1 Efficient knockdown of lamins and nesprins

Linker of nucleoskeleton and cytoskeleton (LINC) complex component nesprin-2 locates in the nuclear envelope to link the actin cytoskeleton and the nuclear lamina, which maitains cellular structure and serves as mechanotransducers between nuclear and extracellular matrix (ECM) [8]. lamin proteins weave a filamentous network in the inner nuclear interior to form lamina meshwork, which provides chromosomal anchoring sites to maintain genome organization. nucleoplasmic lamins bind to chromatin and have been indicated to regulate chromatin accessibility and spatial chromatin organization [21]. However, the contribution of nesprins and lamins to gene expression has not been fully understood.

Herein, we performed an extensive study of how nesprins and lamins regulate gene expression by performing RNA-seq and RT-qPCR when knockdown of nesprins and lamins by inducible shRNA expression. Our results demonstrated that Lamin A could affect RNA synthesis. LMNA could influence chromatin conformation change. SYNE2 (nesprin-2) could modulate chromatin modification (**Figure 1a**). In this work, shScramble, shLaminA, shLMNA, and shSYNE2 were denoted as comparisons between doxycycline-treated groups and untreated groups. Herein, shScramble groups were used to rule out non-specific effects. Doxycycline inducible shRNA expressing cells with scramble shRNA were chosen as control to exclude non-specific targeting effect. A fragment of shRNA that targeting 3' untranslated region (UTR) region in LMNA genes was chosen to knockdown lamin A (shLaminA). A fragment of shRNA that targeting coding sequence (CDS) region in LMNA genes was chosen to knockdown LMNA (shLMNA). A fragment of shRNA that targeting coding sequence (CDS) region in SYNE2 genes was chosen to knockdown nesprin-2 (shSYNE2). The targeting shRNA sequences were listed in Supplementary Table S1.

The relative gene expression and log2 fold change were quantified in doxycycline-treated groups versus untreated groups for RT-qPCR and RNA-seq data, respectively. For RT-qPCR analysis in shScramble comparisons, the mean ± s.d. relative gene expression ratio was 0.96 ± 0.15 for lamin A, 0.95 ± 0.15 for LMNA, 1.03 ± 0.16 for SYNE2; For RNA-seq analysis in shScramble comparisons, the mean log2 fold change was 0.14 ± 0.10 for lamin A, 0.06 ± 0.05 for LMNA, 0.04 ± 0.11 for SYNE2. These data indicated that scramble shRNA did not change gene expression to the target genes. In contrast, targeting shRNAs significantly reduced the corresponding target genes. For RT-qPCR analysis, the mean ± s.d. relative gene expression ratio was 0.33 ± 0.01 for lamin A in shLaminA comparisons, 0.30 ± 0.02 for LMNA in shLMNA comparisons, 0.21 ± 0.04 for SYNE2 in shSYNE2 comparisons; For RNA-seq analysis, the mean log2 fold change was −1.34 ± 0.08 for lamin A in

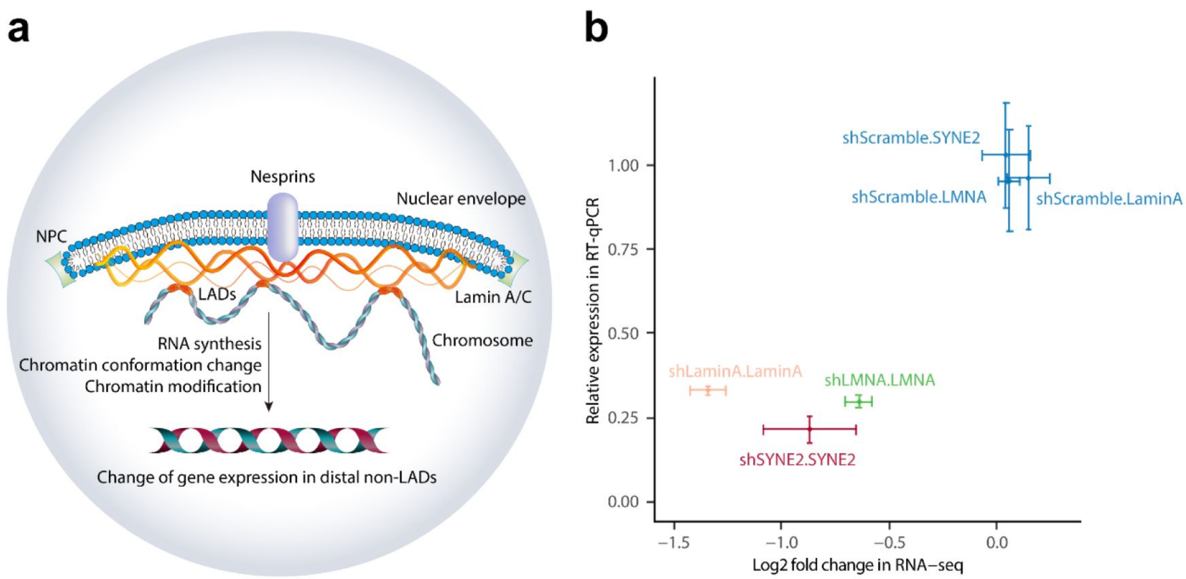

Figure 1.

Knockdown of lamins and nesprins alters gene expression.

a) Schematic illustration of the regulation of gene expression by lamins and nesprins. The lamins and nesprins could affect gene expression in distal non-LADs through altering RNA synthesis, chromatin conformation change, and chromatin modification. NPC, nuclear pore complex; LADs, lamina-associated domains. **b)** Knockdown of lamin A, LMNA, and SYNE2 mRNA levels by targeting shRNAs (shLaminA, yellow; shLMNA, green; shSYNE2, red) was quantified by RT-qPCR and RNA-sequencing (RNA-seq), respectively. In contrast, the relative gene levels of targeting genes (Lamin A, LMNA, SYNE2) are unchanged in the U2OS cells with scramble shRNA construct (shScramble, blue). U2OS cells were treated or untreated with 100 ng ml^{-1} doxycycline for 48 h. The isoform log2 fold change of lamin A was calculated by IsoformSwitchAnalyzeR package. Data represent mean ± s.d. qPCR, n = 7 per group, *p*-value < 0.05; RNA-seq, n = 3 per group, *q*-value < 0.05.

shLaminA comparisons, 0.64 ± 0.06 for LMNA in shLMNA comparisons, −0.87 ± 0.22 for SYNE2 in shSYNE2 comparisons (**Figure 1b**). Herein, we employed IsoformSwitchAnalyzeR to quantify log2 fold change of lamin A isoform (Supplementary Figure S1). Our findings demonstrated a significant reduction in the utilization of the lamin A isoform, while the expression levels of other LMNA gene isoforms remained unchanged. Together, our results indicated the efficient knockdown of lamin A, LMNA and SYNE2 genes. Knockdown of lamin A specifically reduced the usage of its primary isoform, suggesting a potential role in chromatin architecture regulation, while other LMNA isoforms remained unaffected, highlighting a selective effect. Together, our results indicated the efficient knockdown of lamin A, LMNA, and SYNE2 genes.

3.2 Impact of lamins and nesprins on RNA biosynthesis, chromatin conformation, and modification

Correlation heatmap and principal component analysis (PCA) plot revealed distinct gene expression profiles across the shScramble, shLaminA, shLMNA, and shSYNE2 groups (Supplementary Figure S2). Next, by performing DE analysis with DESeq2, our data showed that depletion of lamins and nesprins could significantly lead to the alternation of gene expression, inlcuding the protein-coding mRNAs and long non-coding lncRNAs (Supplementary Table S3). To examine pathway enrichment by the DEGs in the shLaminA, shLMNA, and shSYNE2 comparisons. GO analysis on biological process (BP) was conducted by clusterProfiler package, which was based on the hypergeometric test of desending ranking of log2 fold change of DEGs. No significant BP pathway was enriched for shScramble comparison (Supplementary Figure S3). Intriguingly, RNA synthesis pathways were enriched in the knockdown of lamin A isoform, including purine-containing compound metabolic process and ribose phosphate metabolic process, which ensure the synthesis of the ribonucleotide triphosphates (NTPs) that required for RNA synthesis (**Figure 2a**). In contrast, the knockdown of the LMNA gene was associated with an enrichment of DNA conformation change, suggesting that LMNA may regulate alterations in the spatial organization of chromatin (**Figure 2b**). In addition, the knockdown of the SYNE2 gene was linked to ncRNA metabolic process and covalent chromatin modification, indicating that the impact of SYNE2 in the regulation of chromatin accessibility, remodeling and gene expression (**Figure 2c**).

3.3 Role of lamins and nesprins in isoform switches

To investigate how lamins and nesprins would influence isoform swithes, we performed isoform switches analysis by IsoformSwitchAnalyzeR package. By ascending ranking of q-values of isoform switches, the top 10 isoform genes including TARS1, PPP1R13L, ATP6AP2, NBL1, JADE1, AGPS, Lnc-ZFAT-1, RBM11, and PEX19 were significant changed in the knockdown of lamin A gene (**Figure 3a**). These proteins are involved in protein synthesis (TARS1), chromatin remodeling (JADE1), RNA (RBM11), and protein degration (ATP6AP2, PEX19). The top isoform genes including SAFB2, LIN54, DUXAP9, BCAS3, G6PD, TEAD2, AHR, PAM, and ZFHX4 were significant changed in the knockdown of LMNA gene (**Figure 3b**). These proteins are involved in transcriptional repressors (SAFB2, LIN54), transcription factor (TEAD2, AHR, ZFHX4), and post-translational modification (PAM). The top isoform genes including HNRNPK, ILF3, HNRNPA1, DUXAP9, RPS27, LDHB, CALU, SNHG29, HNRNPDL, and EDC3 were significant changed in the knockdown of LMNA gene (**Figure 3c**). These proteins are involved in RNA splicing and stability (HNRNPK, ILF3, HNRNPA1, HNRNPDL), mRNA decay (EDC3), and protein synthesis and folding (RPS27, RPS27).

Nesprin-2 depletion induced a substantially larger number of DE isoforms (n = 4505) compared to lamin A (n = 1326) and *LMNA* (n = 1156) gene knockdown. This included 645 DE long non-coding lncRNAs and 3860 DE mRNAs in nesprin-2 knockdown cells, as well as 273 DE lncRNAs and 1053 DE mRNAs following lamin A silencing, and 263 DE lncRNAs and 893 DE mRNAs after *LMNA* reduction (**Figure 3d**). While the number of DEGs identified for lamins was limited, a significant number of isoform switches was observed following lamin knockdown.

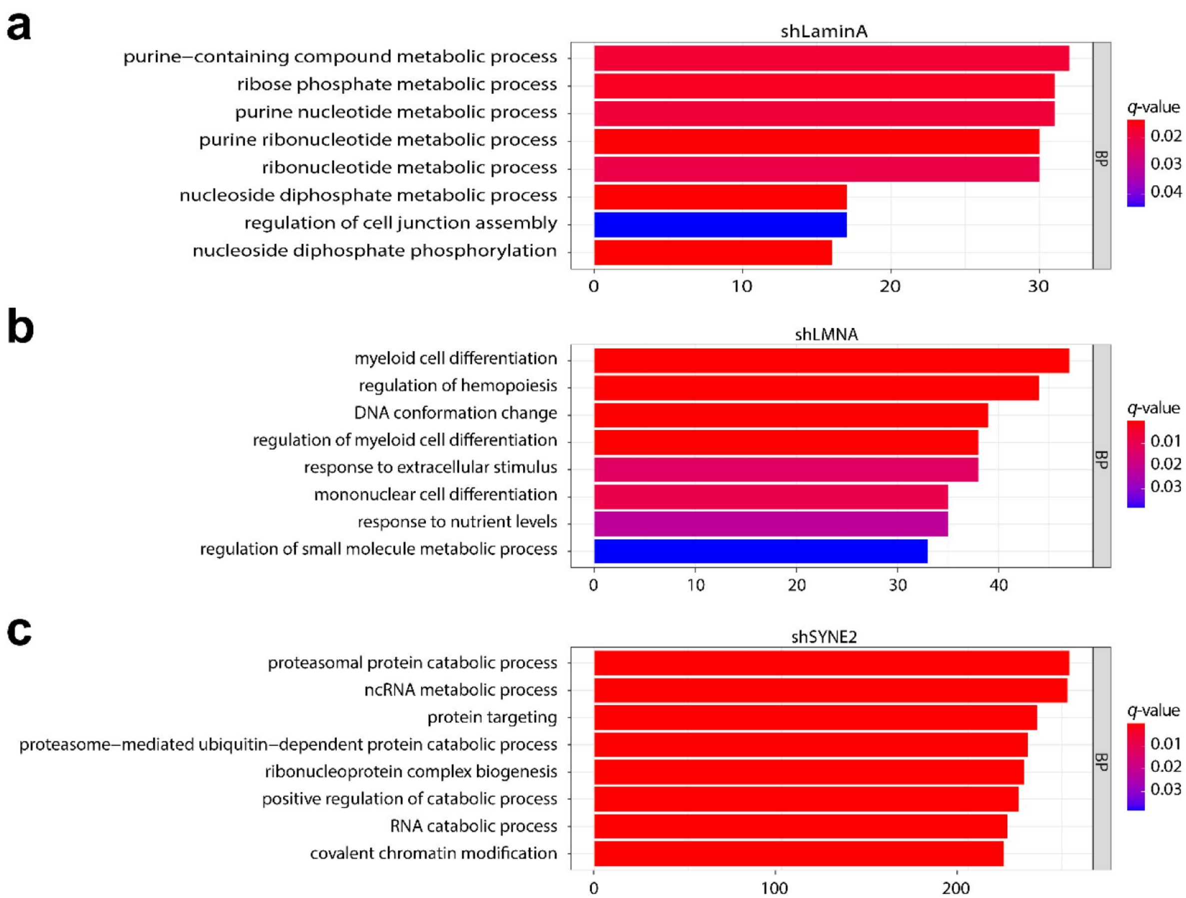

Figure 2.

Impact of Lamins and Nesprins on RNA Biosynthesis, Chromatin Conformation, and Modification.

DEGs were identified using the DESeq2 package by comparing the doxycycline-treated group to the untreated group. Gene Ontology (GO) enrichment analysis of DEGs was conducted using the clusterProfiler package. The top pathways were selected based on a *q*-value cutoff of < 0.05 for shLaminA (**a**), shLMNA (**b**), and shSYNE2 (**c**), respectively.

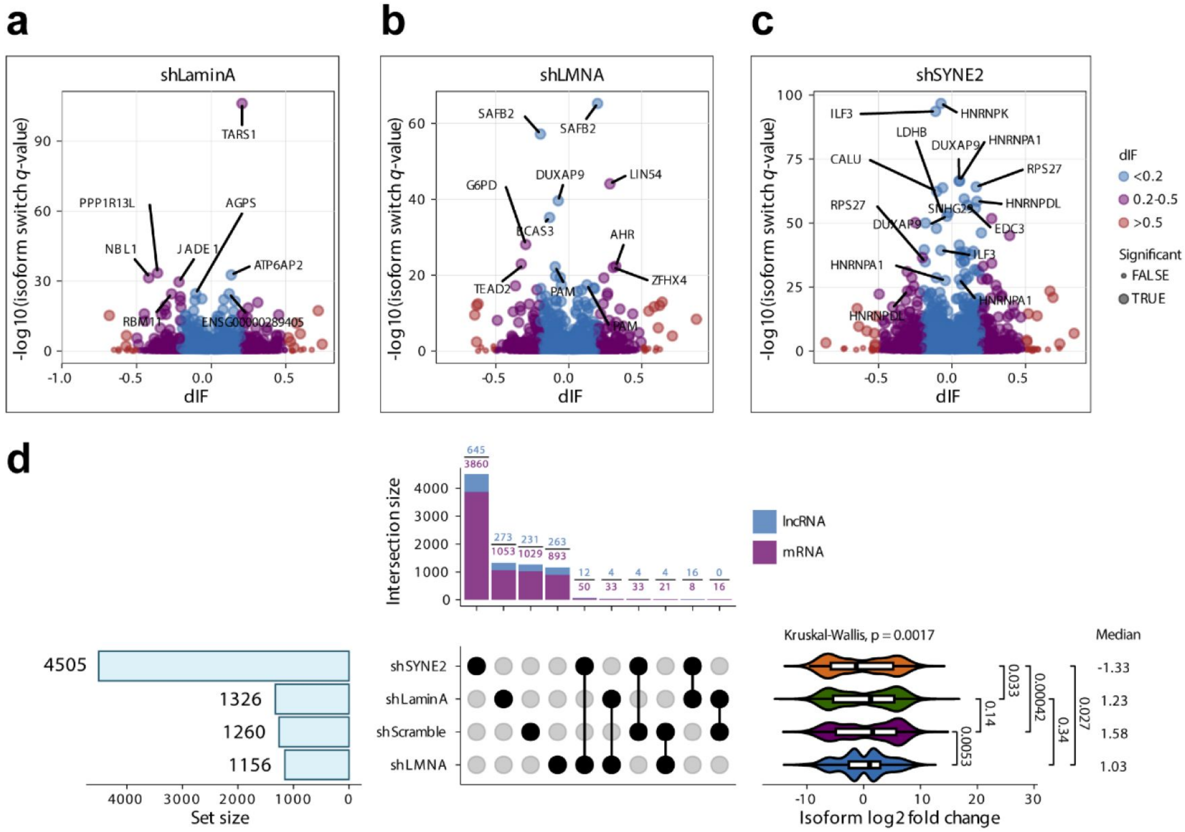

Figure 3.

Roles of Lamins and Nesprins in Isoform Switching.

a, b, c) Volcano plots illustrating the dIF of differentially expressed isoform transcripts against the −log10 of the isoform switch q-values in shLaminA (a), shLMNA (b), and shSYNE2 (c). Isoform switching was detected using IsoformSwitchAnalyzeR (n = 3 per group, cutoff dIF > 0.1, *q*-value < 0.05). a) Lamin A knockdown leads to isoform switching in chromatin remodeling (JADE1) and alternative splicing regulation (RBM11). b) LMNA knockdown results in isoform switching of transcriptional repressors (SAFB2, LIN54) and transcription factors (TEAD2, AHR). c) SYNE2 knockdown significantly alters isoform ratios of RNA-binding proteins (HNRNPK, ILF3, HNRNPA1). d) UpSet plot displaying the number and overlap of differentially expressed protein-coding mRNAs and lncRNAs, as well as the log2 fold change of isoform transcripts in shLaminA, shLMNA, and shSYNE2 conditions. Intersection sizes indicate the number of shared DEGs between knockdowns. Statistical analysis was performed using the non-parametric Kruskal-Wallis test to compare medians across multiple groups.

Interestingly, The UpSet plot analysis revealed a high degree of specificity in the gene expression changes induced by each gene knockdown. Minimal overlaps were observed between the differentially expressed genes in shSYNE2, shLaminA, and shLMNA conditions: shSYNE2 and shLaminA (16 lncRNAs, 8 mRNAs), shSYNE2 and shLMNA (12 lncRNAs, 50 mRNAs), and shLaminA and shLMNA (4 lncRNAs, 33 mRNAs), suggesting distinct downstream effects of each gene knockdown. Relative to the shScramble control (median isoform log2 fold change, 1.58), knockdown of LMNA, LaminA, and SYNE2 resulted in substantial changes in isoform expression, with median log2 fold changes of 1.03, 1.23, and −1.33, respectively. Notably, shSYNE2 induced overall reduction of isoform switches. Together, our results demonstrated that Each gene knockdown from lamin A isoform, LMNA an SYNE2 genes induced distinct and highly specific changes in isoform switches.

3.4 Regulation of gene expression in distal non-LAD regions by lamins and nesprins

The majority of genes located within LADs tend to be transcriptionally repressed or expressed at low levels. This is because LADs are associated with heterochromatin, a tightly packed form of DNA that is generally inaccessible to the cellular machinery required for gene expression. Lamin mutations have shown to disrup LAD organization and gene expression that have been implicated in various diseases, including cancer and laminopathies. Next, we investigated the chromosome localizations of DE mRNAs in the genome-wide. Expectedly, the DEGs from depletion of lamin A, LMNA, and SYNE2 were seldomly intersected with genes in LADs (**Figure 4a**). Consistent with prior studies, depletion of lamin A, LMNA, and SYNE2 did not significantly impact LAD-associated genes, reinforcing the idea that their primary function in gene regulation may occur in distal non-LAD chromatin regions [12].

Interestingly, the DEGs from depletion of lamin A, LMNA, and SYNE2 were dispersed in different chromosome loci when mapping genes to human hg19 genome reference. The DEGs from depletion of lamin A were located in chr8q13, chr10q21, and chr11q25. The DEGs from depletion of LMNA were mainly located in chr20q13, chr3q13, and chr19p12. The DEGs from depletion of SYNE2 were mainly located in chr9q34, chr1p36, and chr17q21 (**Figure 4b**). To confirm that the DEGs were not located in LADs, we measured the spatial distance between LADs and DEGs in shLaminA, shLMNA, and shSYNE2 through GenometriCorr package. Our data suggested that those DEGs are spatially far away from LADs (**Figure 4c**). Together, our results demonstrated that the depletion of lamin A, LMNA, and SYNE2 led to the change of gene expression in distal non-LADs.

3.5 Modulation of lncRNA-mRNA interactions in the nuleus by lamins and nesprins

In accordance to published results, our data also indicated the low expression profiling of lncRNAs compared to mRNA expression profiling (Supplementary Figure S4) [22]. Interestingly, the mRNA expression profiling demonstrated continuously unimodal distribution through density plot of log2 transcript per million (Log2TPM). In contrast, the lncRNA expression profiling demonstrated multimodal dsitribution with multiple peaks (**Figure 5a**). In addition, our predictions showed that lncRNAs are mainly located in nucleus through analysis with DeepLncLoc algorithm (**Figure 5b**) [23]. The results suggested that mRNA-lncRNA interactions were mainly occured in nucleus.

Furthermore, cis-acting mRNA-lncRNA interaction pairs were also obtained by searching highly correlated expression levels between mRNAs and lncRNAs in the10kb windows in the genome-wide (**Figure 5c**). Additionally, the lncTar algorithm and peason correlation test were employed to analyze lncRNA-mRNA coexpression networks, enabling the identification and validation of trans-acting and/or cis-acting lncRNA-mRNA interaction networks [24]. We identified two interaction networks between two lncRNAs (green nodes) and their associated mRNAs (orange nodes). Upregulated genes are represented as triangles, while downregulated genes are shown as

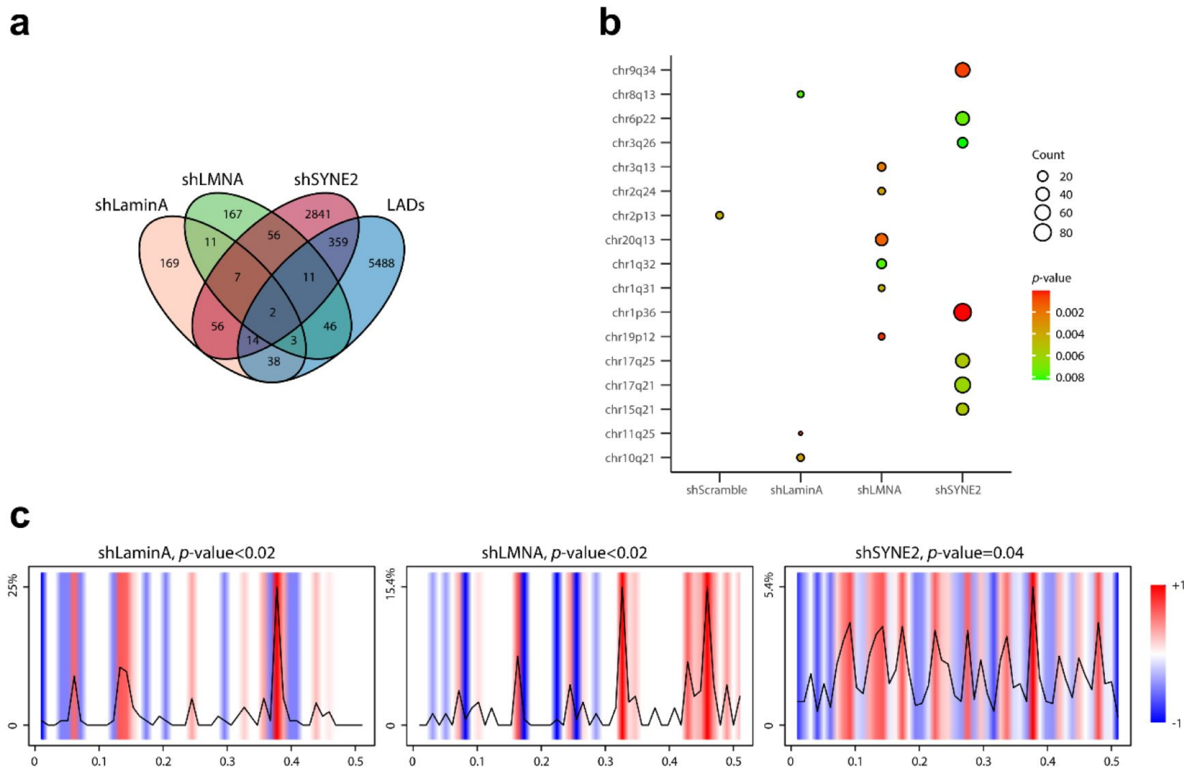

Figure 4.

Regulation of gene expression in distal non-LAD regions by lamins and nesprins.

a) Venn diagram illustrating the number of DE mRNAs in shLaminA (yellow), shLMNA (green), and shSYNE2 (red), with minimal overlap with genes located in LADs (blue). LAD-associated genes were identified using a BED file derived from Lamin A/C ChIP-seq data[28]. **b)** GO enrichment analysis showing the genomic distribution of DEGs across chromosome arms. for shLaminA, shLMNA, and shSYNE2. The *p*-values (blue to red), and the counts of DE genes mapped to the reference chromosome coordinates are shown. Depletion of Lamin A affects chr10q12, LMNA affects chr20q13, and SYNE2 affects chr9q34, chr1q36. **c)** DE genes from shLaminA, shLMNA, and shSYNE2 are spatially distant from LADs. GenometriCorr analysis quantifying spatial distance between LADs and DEGs. The relative distance from DE genes (query features) to LADs (reference feature) is plotted by GenometriCorr package (v 1.1.24). the color depicting deviation from the expected distribution and the line indicating the density of the data at relative distance are shown. Statistical analysis performed using Jaccard test, *p*-value < 0.05.

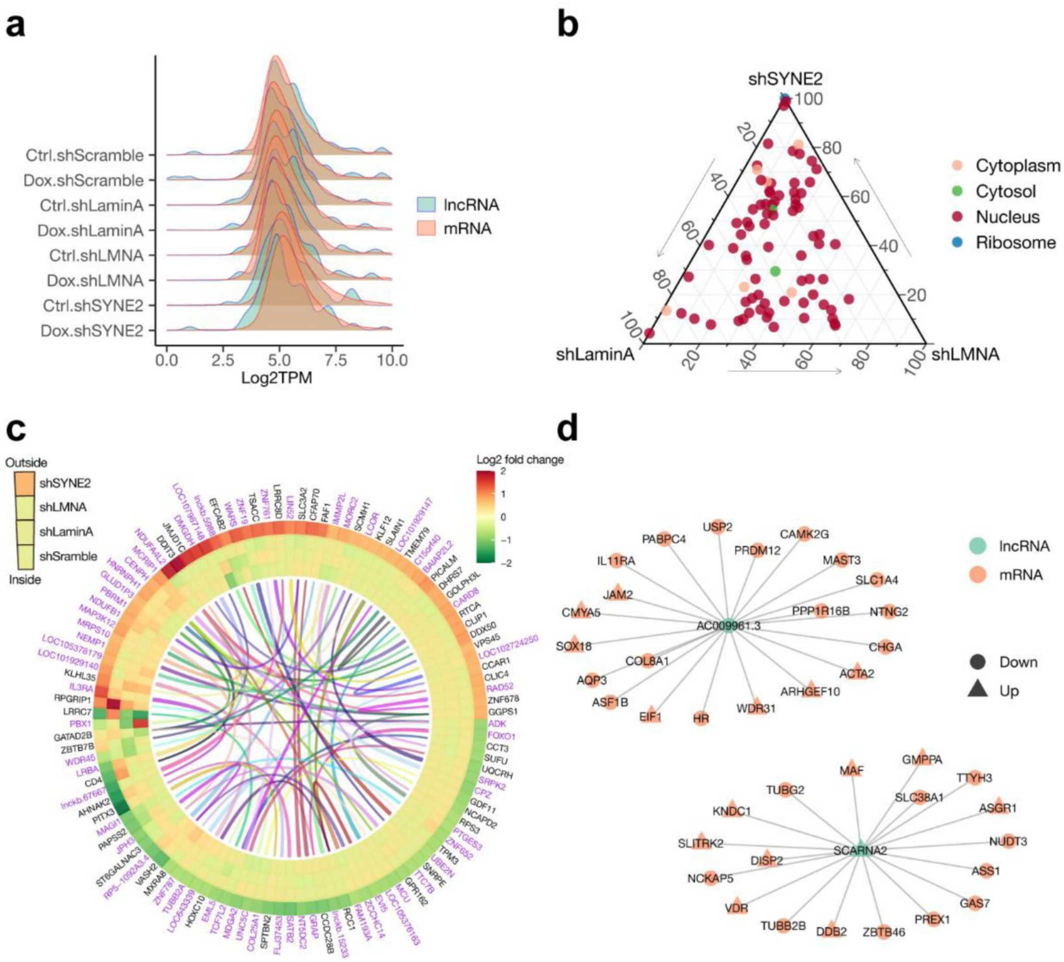

Figure 5.

Modulation of lncRNA-mRNA Interactions by Lamins and Nesprins.

a) Expression distribution of lncRNAs versus mRNAs in log2 transcript per million (Log2TPM), highlighting differences in transcript abundance. **b)** DeepLncLoc predictions showing nuclear localization of identified lncRNAs. **c)** Cis-acting lncRNA-mRNA interaction network predicted based on highly correlated expression levels within 10 kb windows (Pearson correlation > 0.8, q-value < 0.05). **d)** Coexpression network analysis of lncRNA-mRNA interactions. Nodes represent transcripts; edges denote coexpression relationships (Pearson correlation > 0.8, q-value < 0.05).

circles. Two clusters are centered around the lncRNAs AC009961.3 and SCARNA2, highlighting their regulatory roles in specific gene expression pathways. The networks underscored the potential mechanistic influence of lncRNAs in cellular processes and their functional impact on mRNA expression (**Figure 5d**).

3.6 Increased telomere dynamics with lamins and nesprins depletion

To investigate the role of lamins and nesprins in spatial and temporal organization and dynamic behaviors of chromatin, we utilized enhanced green fluorescent protein (EGFP)-tagged dCas9 to visualize telomeric regions of the genome containing repetitive elements in living U2OS cells, allowing us to analyze telomere movement [15]. Representative fluorescence images of telomeres labeled with EGFP-dCas9 in U2OS cells are shown in **Figure 6a**. Notably, depletion of lamin A, LMNA, and SYNE2 resulted in significantly increased genome dynamics and a larger nuclear area scanned. This effect was quantified through telomere mean square displacement (MSD) analysis and the measurement of traveled distance within a 30-second imaging window (**Figure 6b, c** and Supplementary Movie S1). These findings support the concept that lamins play a critical role in regulating chromatin dynamics [25]. Collectively, our results highlight a regulatory function of lamins and nesprins in controlling chromatin behavior.

4. Discussion

LADs collectively account for approximately 40% of the genome, though not all of these regions are bound to the nuclear lamina across all cells within a population. Defined by their physical interaction with the nuclear lamina, LADs are characterized by low gene density—containing approximately 2–3 genes per megabase, in contrast to the human genome's average of 8 genes per megabase. These regions are enriched with repressive chromatin features, limiting transcriptional activity compared to non-LAD regions, which are typically more transcriptionally active [11].

Our findings reveal distinct functional roles for Lamin A, LMNA, and SYNE2 in chromatin regulation and gene expression, providing key insights into the spatial organization of the genome. Notably, Lamin A depletion led to enrichment of pathways associated with RNA biosynthesis, supporting its previously suggested role in transcriptional activation and ribonucleotide metabolism. This aligns with prior studies indicating that Lamin A contributes to chromatin accessibility and RNA polymerase activity. In contrast, LMNA knockdown was linked to alterations in chromatin conformation, underscoring its role in maintaining spatial chromatin organization and nuclear stability. Given that LMNA mutations have been implicated in various laminopathies and genome instability disorders, these findings provide a mechanistic framework linking LMNA function to nuclear integrity.

The findings that DEGs are predominantly located in non-LAD regions highlight a unique regulatory aspect of lamins and nesprins, emphasizing their spatial specificity in gene expression. Moreover, the modulation of lncRNA-mRNA interactions by these proteins unveils a layered regulatory network, further illustrating the nuclear envelope's influence on transcriptional dynamics. The isoform switching patterns observed following knockdown of Lamin A and LMNA provide further insight into transcriptomic flexibility. Our data show that Lamin A knockdown specifically reduced the usage of its primary isoform, suggesting a potential role in chromatin architecture regulation, while other LMNA isoforms remained unaffected, highlighting a selective effect. Given that lamin isoforms are known to differentially impact nuclear stiffness and mechanotransduction, these findings raise the possibility that specific lamin isoforms may be preferentially required for distinct chromatin interactions and nuclear functions.

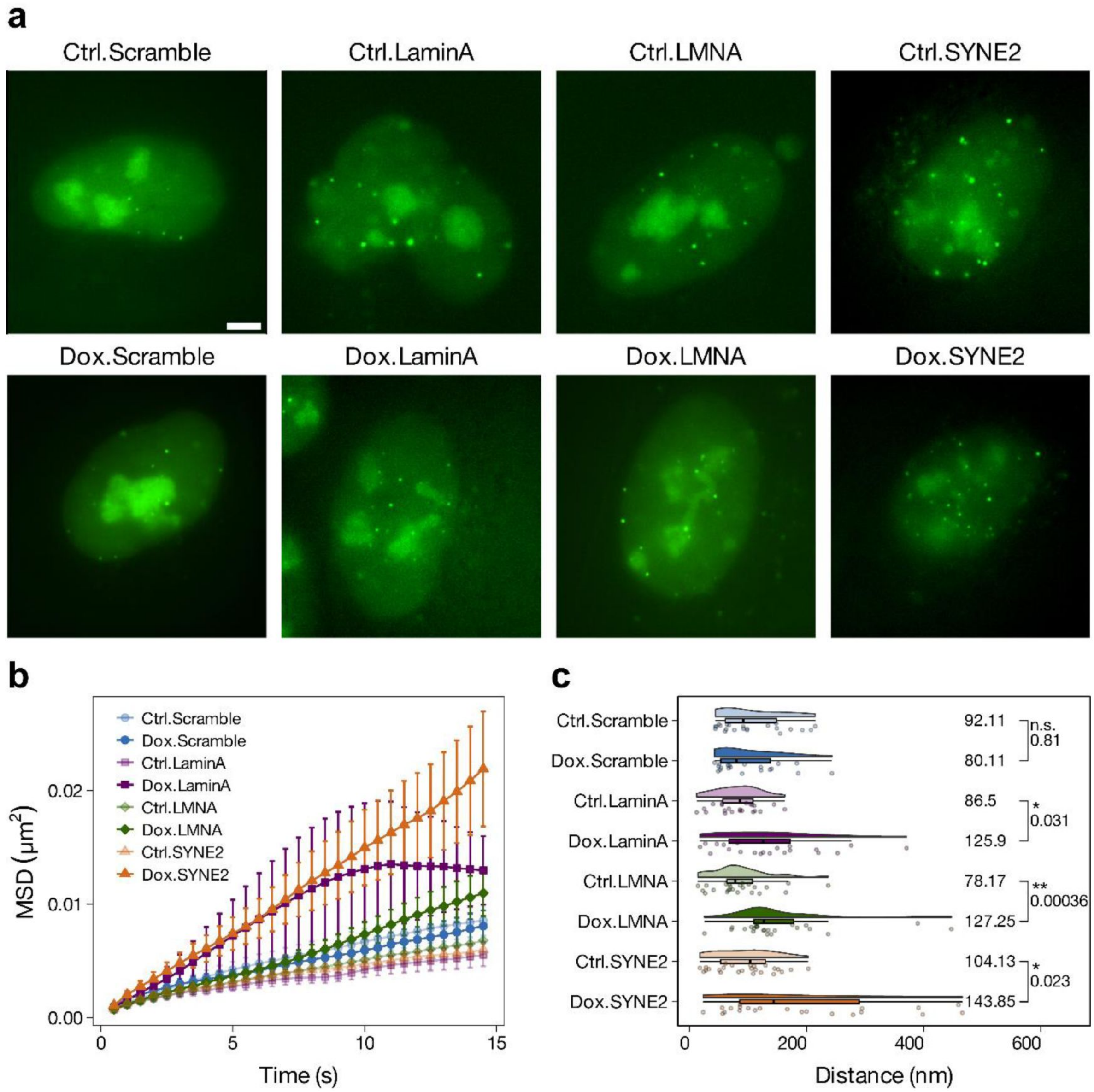

Figure 6.

Increased telomere dynamics following depletion of lamin A, LMNA, and SYNE2.

a) Representative fluorescence images of telomeres labeled with EGFP-dCas9 in U2OS cells. Cells expressing dCas9-EGFP, targeted to telomeric repeats, were imaged via TIRF microscopy. Scale bar = 5 μm. **b)** Mean Square Displacement (MSD) analysis of telomere mobility in control (shScramble) and knockdown conditions (shLaminA, shLMNA, shSYNE2) at 500 ms interval points for 30 s. MSD curves represent n > 20 cells per group. **c)** Total telomere movement range tracked over a 30 s imaging window. Annotated numbers indicate the median values for each group. Statistical analysis was performed using the non-parametric Kruskal-Wallis test, with a significance threshold of *p-value < 0.05, **p-value < 0.001, and n.s. p-value > 0.05.

These results resonate with existing studies linking nuclear envelope dysfunction to diseases such as laminopathies and cancer [4,26,27], where altered nuclear mechanics and chromatin misregulation are hallmarks of disease progression. The observed enhancement of chromatin dynamics upon depletion of Lamin A, LMNA, and SYNE2, as visualized through dCas9-based live chromosome imaging, suggests that these nuclear envelope proteins contribute to chromatin rigidity and nuclear compartmentalization. Their depletion leads to increased chromatin fluidity, which may underlie genomic instability and misregulated transcription in disease states.

By elucidating the molecular mechanisms underlying chromatin behavior, this study contributes to the growing understanding of nuclear architecture as a key regulator of cellular homeostasis. These insights may have broad implications for understanding nuclear envelope-associated diseases, particularly those involving mutations in LMNA or SYNE2, and may help guide future therapeutic strategies targeting nuclear envelope dysfunction.

5. Conclusion

In summary, this study demonstrates the intricate roles of lamin A, LMNA, and SYNE2 in regulating distal chromatin and gene expression. Using inducible knockdown systems and transcriptomic analyses, we identified their distinct contributions to RNA synthesis, chromatin conformation, and modifications. Using dCas9-based live chromosome imaging, enhanced chromatin dynamics were oberserved for the depletion of lamin A, LMNA, and SYNE2 genes. These findings enhance our understanding of nuclear envelope-associated gene regulation and its implications for diseases involving chromatin misregulation. Future research should explore the broader impact of these mechanisms in diverse cellular contexts, advancing potential therapeutic strategies targeting nuclear envelope dysfunction.

Acknowledgements

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Additional information

Compliance and ethics

The authors declare no conflicts of interest or personal relationships influencing this work.

Data availability

BED file were retrieved from lamin A/C CHIP-seq (*https://github.com/kohta-ikegami/pS22-LMNA/blob/master/E1_BJ5ta_LAD.bed*)[28]. All the processed data are provided in the Supplementary Tables. All raw sequencing data produced in this study have been deposited in the NCBI SRA database under accession number PRJNA1212085.

Additional files

Supplementary table S1.

Supplementary movie S1.

Supplementary figures.

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Reviewer #1 (Public review):

This manuscript reports a descriptive study of changes in gene expression after knockdown of the nuclear envelope proteins lamin A/C and Nesprin2/SYNE2 in human U2OS cells. The readout is RNA-seq, which is analyzed at the level of gene ontology and focused investigation of isoform variants and non-coding RNAs. In addition, the mobility of telomeres is studied after these knockdowns, although the rationale in relation to the RNA-seq analyses is rather unclear.

RNA-seq after knockdown of lamin proteins has been reported many times, and the current study does not provide significant new insights that help us to understand how lamins control gene expression. This is particularly because the vast majority of the observed effects on gene expression appear to occur in regions that are not bound by lamin A. It seems likely that these effects are indirect. There is also virtually no overlap between genes affected by laminA/C and by SYNE2, which remains unexplained; for example, it would be good to know whether laminA/C and SYNE2 bind to different genomic regions. The claim in the Title and Abstract that LMNA governs gene expression / acts through chromatin organization appears to be based only on an enrichment of gene ontology terms "DNA conformation change" and "covalent chromatin conformation" in the RNA-seq data. This is a gross over-interpretation, as no experimental data on chromatin conformation are shown in this study. The analyses of transcript isoform switching and ncRNA expression are potentially interesting but lack a mechanistic rationale: why and how would these nuclear envelope proteins regulate these aspects of RNA expression? The effects of lamin A on telomere movements have been reported before; the effects of SYNE2 on telomere mobility are novel (to my knowledge), but should be discussed in the light of previously documented effects of SUN1/2 on the dynamics of dysfunctional telomeres (Lottersberger et al, Cell 2015).

As indicated below, I have substantial concerns about the experimental design of the knockdown experiments.

Altogether, the results presented here are primarily descriptive and do not offer a significant advance in our understanding of the roles of LaminA and SYNE2 in gene regulation or chromatin biology, because the results remain unexplained mechanistically and functionally. Furthermore, the RNAseq datasets should be interpreted with caution until off-target effects of the shRNAs can be ruled out.

Specific comments:

(1) Knockdowns were only monitored by qPCR. Efficiency at the protein level (e.g., Western blots) needs to be determined.

(2) For each knockdown, only a single shRNA was used. shRNAs are infamous for off-target effects; therefore, multiple shRNAs for each protein, or an alternative method such as CRISPR deletion or degron technology, must be tested to rule out such off-target effects.

(3) It is not clear whether the replicate experiments are true biological replicates (i.e., done on different days) or simply parallel dishes of cells done in a single experiment (= technical replicates). The extremely small standard deviations in the RT-qPCR data suggest the latter, which would not be adequate.

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Reviewer #2 (Public review):

Summary:

This study focused on the roles of the nuclear envelope proteins lamin A and C, as well as nesprin-2, encoded by the LMNA and SYNE2 genes, respectively, on gene expression and chromatin mobility. It is motivated by the established role of lamins in tethering heterochromatin to the nuclear periphery in lamina-associated domains (LADs) and modulating chromatin organization. The authors show that depletion of lamin A, lamin A and C, or nesprin-2 results in differential effects of mRNA and lncRNA expression, primarily affecting genes outside established LADs. In addition, the authors used fluorescent dCas9 labeling of telomeric genomic regions combined with live-cell imaging to demonstrate that depletion of either lamin A, lamin A/C, or nesprin-2 increased the mobility of chromatin, suggesting an important role of lamins and nesprin-2 in chromatin dynamics.

Strengths:

The major strength of this study is the detailed characterization of changes in transcript levels and isoforms resulting from depletion of either lamin A, lamin A/C, or nesprin-2 in human osteosarcoma (U2OS) cells. The authors use a variety of advanced tools to demonstrate the effect of protein depletion on specific gene isoforms and to compare the effects on mRNA and lncRNA levels.

The TIRF imaging of dCas9-labeled telomeres allows for high-resolution tracking of multiple telomeres per cell, thus enabling the authors to obtain detailed measurements of the mobility of telomeres within living cells and the effect of lamin A/C or nesprin-2 depletion.

Weaknesses:

Although the findings presented by the authors overall confirm existing knowledge about the ability of lamins A/C and nesprin to broadly affect gene expression, chromatin organization, and chromatin dynamics, the specific interpretation and the conclusions drawn from the data presented in this manuscript are limited by several technical and conceptual challenges.

One major limitation is that the authors only assess the knockdown of their target genes on the mRNA level, where they observe reductions of around 70%. Given that lamins A and C have long half-lives, the effect at the protein level might be even lower. This incomplete and poorly characterized depletion on the protein level makes interpretation of the results difficult. The description for the shRNA targeting the LMNA gene encoding lamins A and C given by the authors is at times difficult to follow and might confuse some readers, as the authors do not clearly indicate which regions of the gene are targeted by the shRNA, and they do not make it obvious that lamin A and C result from alternative splicing of the same LMNA gene. Based on the shRNA sequences provided in the manuscript, one can conclude that the shLaminA shRNA targets the 3' UTR region of the LMNA gene specific to prelamin A (which undergoes posttranslational processing in the cell to yield lamin A). In contrast, the shRNA described by the authors as 'shLMNA' targets a region within the coding sequence of the LMNA gene that is common to both lamin A and C, i.e., the region corresponding to amino acids 122-129 (KKEGDLIA) of lamin A and C. The authors confirm the isoform-specific effect of the shLaminA isoform, although they seem somewhat surprised by it, but do not confirm the effect of the shLMNA construct. Assessing the effect of the knockdown on the protein level would provide more detailed information both on the extent of the actual protein depletion and the effect on specific lamin isoforms. Similarly, given that nesprin-2 has numerous isoforms resulting from alternative splicing and transcription initiation. In the current form of the manuscript, it remains unclear which specific nesprin-2 isoforms were depleted, and to what extent (on the protein level).

Another substantial limitation of the manuscript is that the current analysis, with the exception of the chromatin mobility measurements, is exclusively based on transcriptomic measurements by RNA-seq and qRT-PCR, without any experimental validation of the predicted protein levels or proposed functional consequences. As such, conclusions about the importance of lamin A/C on RNA synthesis and other functions are derived entirely from gene ontology terms and are not sufficiently supported by experimental data. Thus, the true functional consequences of lamin A/C or nesprin depletion remain unclear. Statements included in the manuscript such as "our findings reveal that lamin A is essential for RNA synthesis, ..." (Lines 79-80) are thus either inaccurate or misleading, as the current data do not show that lamin A is ESSENTIAL for RNA synthesis, and lamin A/C and lamin A deficient cells and mice are viable, suggesting that they are capable of RNA synthesis.

Another substantial weakness is that the data and analysis presented in the manuscript raise some concerns about the robustness of the findings. Given that the 'shLMNA' construct is expected to deplete both lamin A and C, i.e., its effect encompasses the depletion of lamin A, which is achieved by the 'shLaminA' construct, one would expect a substantial overlap between the DEGs in the shLMNA and shLaminA conditions, with the shLMNA depletion producing a broader effect as it targets both lamin A and C. However, the Venn Diagram in Figure 4a, the genomic loci distribution in Figure 4b, and the correlation analysis in Supplementary Figure S2 show little overlap between the shLMNA and shLaminA conditions, which is quite surprising. In the mapping of the DEGs shown in Figure 4b, it is also surprising not to see the gene targeted by the shRNA, LMNA, found on chromosome 1, in the results for the shLMNA and shLamin A depletion.

The correlation analysis in Supplementary Figure S2 raises further questions. The authors use doc-inducible shRNA constructs to target lamin A (shLaminA), lamin A/C (shLMNA), or nesprin-2 (shSYNE2). Thus, the no-dox control (Ctr) for each of these constructs would be expected to be very similar to the non-target scrambled controls (Ctrl.shScramble and

Dox.shScramble). However, in the correlation matrix, each of the no-dox controls clusters more closely with the corresponding dox-induced shRNA condition than with the Ctrl.shScramble or Dox.shScramble conditions, suggesting either a very leaky dox-inducible system, strong effects from clonal selection, or substantial batch effects in the processing. Either of these scenarios could substantially affect the interpretation of the findings. For example, differences between different clonal cell lines used for the studies, independent of the targeted gene, could explain the limited overlap between the different shRNA constructs and result in apparent differences when comparing these clones to the scrambled controls, which were derived from different clones.

The manuscript also contains several factually inaccurate or incorrect statements or depictions. For example, the depiction of the nuclear envelope in Figure 1 shows a single bilipid layer, instead of the actual double bi-lipid layer of the inner and outer nuclear membranes that span the nuclear lumen. The depiction further lacks SUN domain proteins, which, together with nesprins, form the LINC complex essential to transmit forces across the nuclear envelope. The statement in line 214 that "Linker of nucleoskeleton and cytoskeleton (LINC) complex component nesprin-2 locates in the nuclear envelope to link the actin cytoskeleton and the nuclear lamina" is not quite accurate, as nesprin-2 also links to microtubules via dynein and kinesin.

The statement that "Our data show that Lamin A knockdown specifically reduced the usage of its primary isoform, suggesting a potential role in chromatin architecture regulation, while other LMNA isoforms remained unaffected, highlighting a selective effect" (lines 407-409) is confusing, as the 'shLaminA' shRNA specifically targets the 3' UTR of lamin A that is not present in the other isoforms. Thus, the observed effect is entirely consistent with the shRNA-mediated depletion, independent of any effects on chromatin architecture.

The premise of the authors that lamins would only affect peripheral chromatin and genes at LADs neglects the fact that lamins A and C are also found in the nuclear interior, where they form stable structure and influence chromatin organization, and the fact that lamins A and C and nesprins additionally interact with numerous transcriptional regulators such as Rb, c-Fos, and beta-catenins, which could further modulate gene expression when lamins or nesprins are depleted.

The comparison of the identified DEGs to genes contained in LADs might be confounded by the fact that the authors relied on the identification of LADs from a previous study (ref #28), which used a different human cell type (human skin fibroblasts) instead of the U2OS osteosarcoma cells used in the present study. As LADs are often highly cell-type specific, the use of the fibroblast data set could lead to substantial differences in LADs.

Another limitation of the current manuscript is that, in the current form, some of the figures and results depicted in the figures are difficult to interpret for a reader not deeply familiar with the techniques, based in part on the insufficient labeling and figure legends. This applies, for example, to the isoform use analysis shown in Figure 3d or the GenometriCorr analysis quantifying spatial distance between LADs and DEGs shown in Figure 4c.

Overall appraisal and context:

Despite its limitations, the present study further illustrates the important roles the nuclear envelope proteins lamin A, lamin C, and nesprin-2 have in chromatin organization, dynamics, and gene expression. It thus confirms results from previous studies (not always fully acknowledged in the current manuscript) previously reported for lamin A/C depletion. For example, the effect of lamin A/C depletion on increasing mobility of chromatin had already been demonstrated by several other groups, such as Bronshtein et al. Nature Comm 2015 (PMID: 26299252) and Ranade et al. BMC Mol Cel Biol 2019 (PMID: 31117946). Additionally, the effect of lamin A/C depletion on gene and protein expression has already been extensively studied in a variety of other cell lines and model systems, including detailed proteomic studies (PMIDs 23990565 and 35896617).

The finding that that lamin A/C or nesprin depletion not only affects genes at the nuclear periphery but also the nuclear interior is not particularly surprising giving the previous studies and the fact that lamins A and C are also founding within the nuclear interior, where they affect chromatin organization and dynamics, and that lamins A/C and nesprins directly interact with numerous transcriptional regulators that could further affect gene expression independent from their role in chromatin organization.

The authors provide a detailed analysis of isoform switching in response to lamin A/C or nesprin depletion, but the underlying mechanism remains unclear. Similarly, their analysis of the genomic location of the observed DEGs shows the wide-ranging effects of lamin A/C or nesprin depletion, but lets the reader wonder how these effects are mediated. A more in-depth analysis of predicted regulator factors and their potential interaction with lamins A/C or nesprin would be beneficial in gaining more mechanistic insights.

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Reviewer #3 (Public review):

Summary:

This manuscript describes DOX inducible RNAi KD of Lamin A, LMNA coded isoforms as a group, and the LINC component SYNE2. The authors report on differentially expressed genes, on differentially expressed isoforms, on the large numbers of differentially expressed genes that are in iLADs rather than LADs, and on telomere mobility changes induced by 2 of the 3 knockdowns.

Strengths:

Overall, the manuscript might be useful as a description for reference data sets that could be of value to the community.

Weaknesses:

The results are presented as a type of data description without formulation of models or explanations of the questions being asked and without follow-up. Thus, conceptually, the manuscript doesn't appear to break new ground.

Not discussed is the previous extensive work by others on the nucleoplasmic forms of LMNA isoforms. Also not discussed are similar experiments- for instance, gene expression changes others have seen after lamin A knockdowns or knockouts, or the effect of lamina on chromatin mobility, including telomere mobility - see, for example, a review by Roland Foisner (doi.org/10.1242/jcs.203430) on nucleoplasmic lamina. The authors need to do a thorough search of the literature and compare their results as much as possible with previous work.

The authors don't seem to make any attempt to explore the correlation of their findings with any of the previous data or correlate their observed differential gene expression with other epigenetic and chromatin features. There is no attempt to explore the direction of changes in gene expression with changes in nuclear positioning or to ask whether the genes affected are those that interact with nucleoplasmic pools of LMNA isoforms. The authors speculate that the DEG might be related to changing mechanical properties of the cells, but do not develop that further.

The technical concerns include: 1) Use of only one shRNA per target. Use of additional shRNAs would have reduced concern about possible off-target knockdown of other genes; 2) Use of only one cell clone per inducible shRNA construct. Here, the concern is that some of the observed changes with shRNA KDs might show clonal effects, particularly given that the cell line used is aneuploid. 3) Use of a single, "scrambled" control shRNA rather than a true scrambled shRNA for each target shRNA.

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